Timing-Only Channel Estimation and Equalization for OFDM Through a Dispersive Time-Modulated RIS

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***Abstract*—A time-modulated reconfigurable intelligent surface (TM-RIS) switches its elements several times within each OFDM symbol. The switching couples every received subcarrier to many transmitted ones, and with frequency-dispersive elements the coupling matrix is neither diagonal nor Toeplitz. Existing TM-RIS designs treat this coupling as interference, which caps the legitimate user's signal-to-interference ratio at -4.8 dB for a 75% duty cycle, and the receiver for the dispersive model was left open. We show that grouping the harmonic expansion by element state separates the coupling exactly into Toeplitz factors fixed by the switching timing, which the receiver knows, and per-state input responses that absorb the propagation *and* the element dispersion, which it does not. From this separation we build, to our knowledge, the first pilot-based receiver for this channel: a low-rank estimator that needs no element model, a blind estimator of the RIS clock offset that rests on the fact that an offset is exactly an opposite delay of the two hops, and an exact matrix-free equalizer. In pre-registered tests with 30 channels per point, the receiver comes within 0.1 bit/s/Hz of perfect channel knowledge with two pilot symbols on a shared-path channel and needs four to eight on 3GPP TDL-C. At 20 and 30 dB it beats every estimator that, like it, needs no training data, on every channel and pilot count: an ideal-element estimator with the same prior and basis-expansion estimators with up to all harmonics and 32 pilot symbols. The ideal-element estimator matches ours when the elements are ideal and degrades as they become less ideal, while ours does not. An LMMSE whose prior is learned from 300 channels of the test distribution is better than ours, but trained on a different delay spread it is worse by up to 0.95 bit/s/Hz; a prior learned from our own 8-pilot estimates, with no true channels and no element model, keeps most of the learned gain. Our receiver decodes coded blocks within 0.14 dB of perfect knowledge, and it absorbs physical clock offsets with no significant rate loss.**

***Index Terms*—Reconfigurable intelligent surface, time modulation, OFDM, intercarrier interference, channel estimation, equalization.**

I. Introduction

A time-modulated RIS (TM-RIS), or space-time-coding metasurface, switches each element several times within an OFDM symbol [1]. The reflected signal then carries harmonics of the switching rate, which has been used for waveform security and directional modulation [2], [3]. For the user the surface serves, the same harmonics move energy from transmitted subcarrier m to received subcarrier $m + h$, so the channel matrix is not diagonal. Iudice *et al.* [4] recently derived a physically consistent model of this channel with frequency-dispersive (finite-memory) elements, as real varactor cells are. They showed that the coupling survives any cyclic prefix, that with dispersive elements the coupling matrix is not even Toeplitz. They assume the channel known and name joint subcarrier equalization as future work [4, Sec. V]; channel estimation is not treated. Prior TM-RIS designs keep a one-tap receiver for the legitimate user and treat the coupling as interference [2], [3]. A blind eavesdropper attack [5] relies on the Toeplitz structure of ideal, frequency-flat switching, which dispersion breaks. Generic estimators of doubly selective OFDM channels, such as the basis-expansion model (CE-BEM) [6], do not know where the coupling comes from: they must estimate every harmonic. Because rectangular switching leaves a $1/h^2$ energy tail, they either truncate the harmonics, which caps their accuracy, or keep all of them, which needs many pilots.

Novelty. The receiver of a TM-RIS link knows *how* the channel varies inside the symbol: the variation is the switching pattern, which the network sets. It does not know the propagation or the exact element response. We turn this observation into an exact structural statement and build the whole receiver on it. Our contributions are:

1) *Timing–propagation separation* (Proposition 1). Grouping the expansion of [4, eq. (5)] by element state gives $\boldsymbol{H} = \sum_{q,s} \boldsymbol{T}_{q,s} \circ (\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}})$. The Toeplitz matrices $\boldsymbol{T}_{q,s}$ depend only on the switching timing. The element dispersion enters only the per-state input response $\boldsymbol{v}_{q,s}$, together with the unknown BS–RIS channel. The regrouping is elementary and we credit it to the anchor model; what is new is what it implies for a receiver.
2) *A receiver that needs no element model*: a low-rank, shared-subspace estimator of $\boldsymbol{H}$ that treats $\boldsymbol{v}_{q,s}$ as unknown (Algorithm 1). Unlike [5], it does not rely on Toeplitz structure; unlike CE-BEM, it never estimates harmonics separately.
3) *Blind clock-offset recovery.* We prove that an offset between the RIS and the receiver is exactly an opposite delay of the two hops (Proposition 2), and we identify the offset physically (Proposition 3). This explains why a wrong offset overfits the pilots, and it leads to an estimator of the offset and the delay reference by Bayesian evidence (Algorithm 2).
4) *A pre-registered evaluation*: pass/fail criteria fixed before

each run, 30 channels per point, paired confidence intervals, settings chosen on channels disjoint from the reported ones, the strongest model-based and learned baselines we could build, and failures reported, including where a learned prior beats ours.

We adopt and do not claim: the DPSS delay basis [7], [8], low-rank RIS channel models [9], matrix-free iterative equalization [10] (exact here rather than a BEM approximation), and evidence-based model selection [11]. The closest timing work, blind offset estimation for RIS symbiotic radio [12], estimates the offset of a backscattered data stream from virtual subcarriers; our receiver instead uses the known pattern and needs no idle subcarriers.

Notation: $\circ$ and $\odot$ denote Hadamard products of matrices and vectors; $(\cdot)^{\mathsf{T}}$, $(\cdot)^{\mathsf{H}}$ and $(\cdot)^*$ denote transpose, conjugate transpose and conjugate; $\operatorname{sinc} x = \sin(\pi x)/(\pi x)$; $\mathcal{CN}$ denotes the circular complex Gaussian distribution.

II. System Model

A base station (BS) transmits OFDM with M subcarriers $f_m = f_c + m\Delta f$, $m \in \mathcal{M} = \{-M/2, \dots, M/2-1\}$, useful duration $T_u = 1/\Delta f$ and cyclic prefix (CP) T_{cp}, to a user through a RIS of Q elements (Fig. 1). Element q follows a periodic pattern $s_q[k] \in \{1, 2\}$, $k = 0, \dots, K-1$, of period T_u: it is in state $s_q[k]$ during slot $[kT_u/K, (k+1)T_u/K)$. The duty cycle D is the fraction of slots in state 1. In state s the element reflects with the single-resonance response of [4],

$$\Gamma_s(f) = -1 + \frac{2\xi_{r,s}}{j2\pi(f - f_{0,s}) + \xi_{r,s} + \xi_{i,s}}, \tag{1}$$

which for the published OpenRIS cell ($f_{0,1} = 3.471\,\mathrm{GHz}$, $f_{0,2} = 3.710\,\mathrm{GHz}$, $\xi_{r,s}/2\pi = 97.5$, $128.4\,\mathrm{MHz}$, $\xi_{i,s}/2\pi = 8.9$, $9.0\,\mathrm{MHz}$ [4, Tab. I]) reproduces the in-channel behaviour reported there (Fig. 2): across the 100 MHz channel at $f_c = 3.594\,\mathrm{GHz}$ each state drifts by 46° and 49° in phase, the two states differ by $180^\circ \pm 7.8^\circ$ with up to 0.57 dB magnitude imbalance, and the reflection loss reaches 1.0 dB. The pair is thus a good binary phase shifter but not the frequency-flat, lossless ± 1 switch of earlier work. Over the 7.7 MHz band of our $M = 256$ comparisons the departure is mainly a static one (a phase difference of 172.2°–172.5°); over the full band the dispersion matters as well. Let $a_q^{\mathrm{s}}(f)$ and $a_q^{\mathrm{d}}(f)$ be the BS–element and element–user responses. When the CP covers the cascaded delay spread plus the element memory (13.8 ns), the received frequency-domain symbol is [4]

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}, \quad H_{\bar{m}m} = \sum_{q=1}^{Q} a_q^{\mathrm{d}}(f_{\bar{m}})\, b_q^{[\bar{m}-m]}(f_m)\, a_q^{\mathrm{s}}(f_m), \tag{2}$$

where $\boldsymbol{w} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ and $b_q^{[h]}(f)$ is the h-th Fourier coefficient, over one period, of the switched response of element q at input frequency f. Equation (2) is the ground truth of our evaluation. Its implementation agrees with an independent time-domain simulation (filter, gate, FFT, with no harmonic algebra) to 0.27% median relative error.

TABLE I
Assumptions, justification, and where they are tested.

Assumption	Justification; test of its violation
A1: pattern $s_q[k]$ known	The network drives the RIS controller, so the pattern is control information, like a pilot configuration. Without it the coupling cannot be told apart from propagation: a receiver that ignores one of two panels collapses to 1.15 bit/s/Hz (Sec. V-G).
A2: generalized CP	Condition of [4]. Checked on the cascade: 3GPP TDL at 30 and 100 ns is valid (≤ -56 dB of energy beyond the CP); 300 ns is not and is out of scope.
A3: period T_u, aligned up to an offset d	TM-IRS designs synchronize the pattern to the symbol [2]; the residual offset is estimated (Sec. IV-C).
A4: shared paths	A compact surface sees the same paths; this is what makes the factors low-rank. Tested with 3GPP TDL taps from random directions, spatially consistent CDL, 1024 elements and a near-field user.
A5: channel constant over a refresh block	The coherence time $0.423/f_D$ [13] spans 1188, 119 and 30 symbols at 3, 30 and 120 km/h. Mobility is tested with the refresh interval selected under pilot overhead (Sec. V-F).
A6: σ^2 known; no mutual coupling	σ^2 is estimated from empty resource elements. Mutual coupling is ignored, as in [4]; this is a stated limitation.

Assumptions. Table I lists each assumption with its justification and where its violation is evaluated. The receiver does *not* assume (1): it only defines the truth.

III. Structure of the Coupling

Proposition 1 (Timing–propagation separation)**.** *For a K-slot pattern of period T_u, $b_q^{[h]}(f) = \sum_{s=1}^{2} t_{q,s}[h]\Gamma_s(f)$ with*

$$t_{q,s}[h] = \frac{e^{-j\pi h/K}}{K} \operatorname{sinc}\left(\frac{h}{K}\right) \sum_{k:\, s_q[k]=s} e^{-j2\pi hk/K}. \tag{3}$$

Hence, with $[\boldsymbol{T}_{q,s}]_{\bar{m}m} = t_{q,s}[\bar{m} - m]$ (Toeplitz), $\boldsymbol{a}_q^{\mathrm{d}} = [a_q^{\mathrm{d}}(f_m)]_m$ and $\boldsymbol{v}_{q,s} = \boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}$,

$$\boldsymbol{H} = \sum_{q=1}^{Q} \sum_{s=1}^{2} \boldsymbol{T}_{q,s} \circ \left(\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}}\right). \tag{4}$$

Proof. In slot k the element applies $\Gamma_{s_q[k]}(f)$, so its switched response is $\sum_s \Gamma_s(f) \mathbb{1}_s(t)$, where $\mathbb{1}_s$ indicates the slots in state s. The h-th Fourier coefficient of the indicator of slot k is $\frac{1}{T_u}\int_{kT_u/K}^{(k+1)T_u/K} e^{-j2\pi ht/T_u}dt = \frac{1}{K}e^{-j\pi h/K}\operatorname{sinc}(h/K)e^{-j2\pi hk/K}$. Summing over the slots in state s gives (3). Substituting into (2), $t_{q,s}$ depends on $\bar{m} - m$ only, which gives (4). □

Fig. 3 shows the resulting structure: the main diagonal is the zero-order reflection and the off-diagonals are the harmonics, whose envelope follows $|\operatorname{sinc}(h/K)|$; for $K = 2$ the even harmonics vanish, which our implementation reproduces from [4, Fig. 5].

Proposition 1 has five consequences, which together define the receiver.

Corollary 1 (No element model)**.** *The element response enters* (4) *only through $\boldsymbol{v}_{q,s}$, which also contains the unknown*

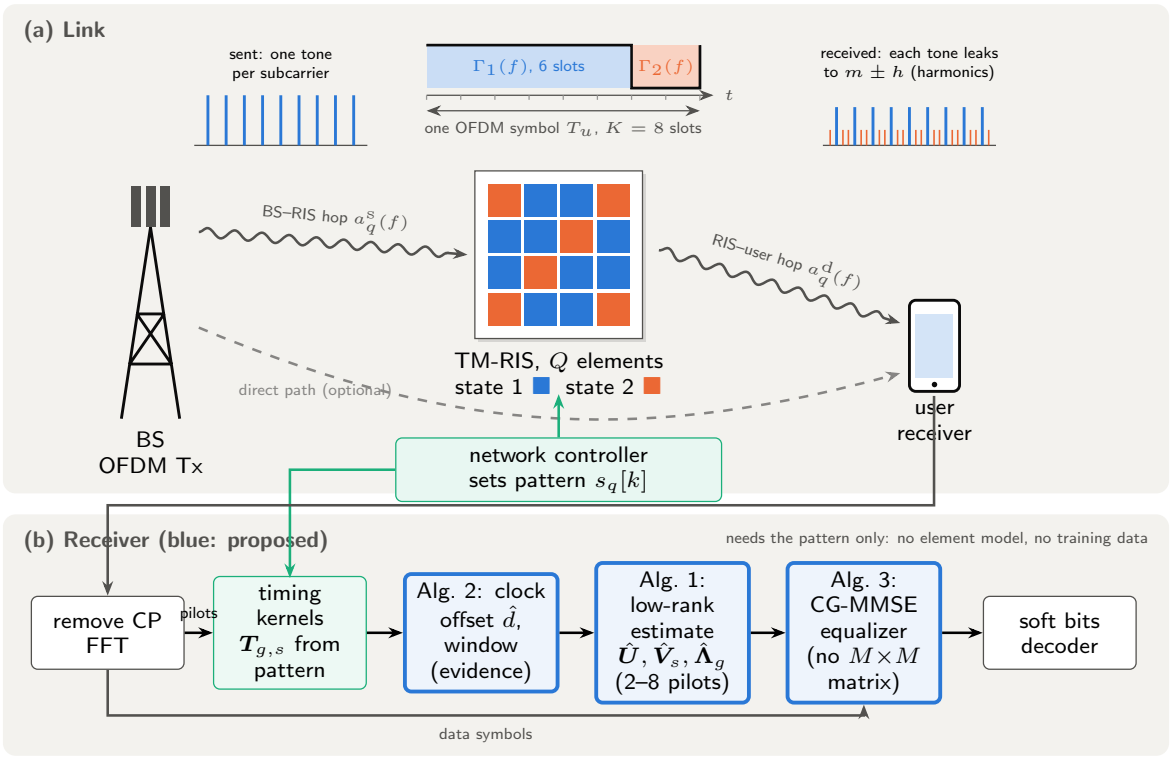


Fig. 1. (a) The link: each RIS element switches between two states within every OFDM symbol (here $K = 8$ slots, duty cycle $D = 0.75$), so every transmitted subcarrier reaches the user spread over harmonic offsets. (b) The receiver: it uses the switching pattern from the network to build the timing kernels, never the element model, and it never forms the $M \times M$ coupling matrix. Blue: the proposed algorithms.

BS–RIS channel. Estimating $\boldsymbol{v}_{q,s}$ as an unknown absorbs the dispersion exactly; the receiver needs only the timing kernels $\boldsymbol{T}_{q,s}$.

Corollary 2 (One-tap ceiling)**.** *For ideal ± 1 elements with a common pattern of duty D, $b^{[0]} = t_1[0] - t_2[0] = 2D - 1$, and Parseval's theorem gives $\sum_h |b^{[h]}|^2 = \frac{1}{T_u}\int_0^{T_u} |\pm 1|^2 dt = 1$. A one-tap receiver collects the fraction $(2D-1)^2$ of the reflected energy and sees the rest as interference, so its SIR is at most $(2D-1)^2/(1-(2D-1)^2)$: $-4.8\,dB$ at $D = 0.75$. Joint equalization collects all of it.*

Corollary 3 (Harmonic tail)**.** *From* (3), $|t_{q,s}[h]| \le |\operatorname{sinc}(h/K)|$, *which decays as $1/|h|$, so the energy outside $|h| \le H$ decays as $1/H$. For $K = 8$ and $D = 0.75$ it is 3.4%, 1.0% and 0.32% for $H = 12$, 40 and 128 (Fig. 3c). A model truncated to $|h| \le 12$ therefore has a normalized error floor of $10\log_{10} 0.034 = -14.7\,dB$, whatever its estimator.*

Corollary 4 (Fast products)**.** *$(\boldsymbol{T} \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}))\boldsymbol{x} = \boldsymbol{a} \odot (\boldsymbol{T}(\boldsymbol{b} \odot \boldsymbol{x}))$, and $\boldsymbol{T}\boldsymbol{z}$ is a linear convolution. Hence $\boldsymbol{H}\boldsymbol{x}$ and $\boldsymbol{H}^{\mathsf{H}}\boldsymbol{y}$ cost $\mathcal{O}(QM \log M)$ with FFTs of length at least $3M - 2$, and $\boldsymbol{H}$ is never formed.*

Proposition 2 (An offset is an opposite delay)**.** *If the switching runs d slots late relative to the receiver's timing, each kernel becomes $\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*$ with $\boldsymbol{D} = \operatorname{diag}(e^{-j2\pi md/K})_{m \in \mathcal{M}}$, and*

$$(\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*) \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}) = \boldsymbol{T} \circ \big((\boldsymbol{D}\boldsymbol{a})(\boldsymbol{D}^*\boldsymbol{b})^{\mathsf{T}}\big). \tag{5}$$

Multiplying a frequency response by $\boldsymbol{D}$ delays it by dT_u/K; multiplying it by $\boldsymbol{D}^*$ advances it by the same amount.

Proof. A shift of d slots multiplies the h-th coefficient by $e^{-j2\pi hd/K} = e^{-j2\pi \bar{m}d/K} e^{j2\pi md/K}$ for $h = \bar{m} - m$. This is (5) entrywise. □

Proposition 3 (Physical timing)**.** *Let the two hops have bulk delays τ_b and τ_u, and let the receiver synchronize to the first cascaded arrival. (i) If the RIS controller is locked to the signal arriving at the RIS, the synchronized $\boldsymbol{H}$ equals the one without bulk delays. (ii) If it is locked to the BS clock, the synchronized $\boldsymbol{H}$ equals the one without bulk delays and with offset $d = -\tau_b K/T_u$.*

Proof. The delays multiply the input side by $e^{-j2\pi f_m \tau_b}$ and the output side by $e^{-j2\pi f_{\bar{m}} \tau_u}$; synchronization multiplies the output by $e^{j2\pi f_{\bar{m}}(\tau_b + \tau_u)}$. In (i) the switching edges are delayed by τ_b, which by Proposition 2 contributes $e^{-j2\pi(\bar{m}-m)\Delta f \tau_b}$, and all phases cancel. In (ii) the edges are not delayed, and what remains is the phase of an offset of $-\tau_b$. □

Both identities hold numerically to $7 \cdot 10^{-15}$. A practical RIS therefore sees either no offset or an offset equal to its BS distance, tens to hundreds of nanoseconds; the latter costs 1.6/4.2 bit/s/Hz at 20/30 dB if ignored (Table IV).

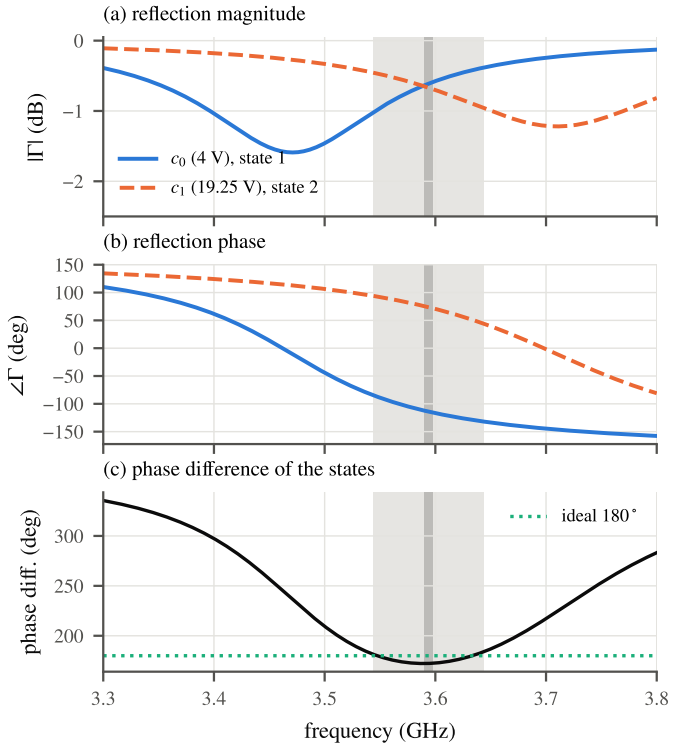


Fig. 2. The OpenRIS cell in the format of [4, Figs. 3–4]: single-pole model (1) with the parameters and common phase -10.7° of [4, Tab. I]. Light band: the 100 MHz channel; dark strip: the 7.7 MHz band of the $M = 256$ comparisons.

IV. The Receiver

A. Low-rank model with a delay prior

Under A4, the user-hop response of element q is $a_q^{\mathrm{d}}(f) = \sum_\ell g_\ell e^{-j\boldsymbol{k}\boldsymbol{u}_\ell^{\mathsf{T}}\boldsymbol{r}_q} e^{-j2\pi f\tau_\ell}$, so $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{E}\boldsymbol{\alpha}_q$ with $\boldsymbol{E} = [e^{-j2\pi f_m\tau_\ell}]_{m,\ell}$ common to all elements. Likewise $\boldsymbol{v}_{q,s} = \mathrm{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'\boldsymbol{\beta}_q$, where $\boldsymbol{\beta}_q$ (the input-hop path gains with the element's geometric phases) is common to both states. Hence $[\boldsymbol{a}_1^{\mathrm{d}}, \dots, \boldsymbol{a}_Q^{\mathrm{d}}]$ has rank at most the number of paths, whatever Q. Every hop response is delay-limited to a window $[t_0, t_0 + \tau_{\max}]$, where $\tau_{\max}$ also covers the element memory. We represent the responses with the first L discrete prolate spheroidal sequences (DPSS) [7], [8]. These are the columns of $\boldsymbol{F} \in \mathbb{C}^{M\times L}$: the leading eigenvectors of $R_{nm} = e^{-j\pi(n-m)\Delta f\tau_{\max}}\,\mathrm{sinc}((n-m)\Delta f\tau_{\max})$, the frequency correlation of a response with delays uniform in the window. Their normalized eigenvalues $\boldsymbol{\lambda}$ serve as prior variances. Compressing the path gains to rank r gives $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{F}\boldsymbol{U}\tilde{\boldsymbol{\alpha}}_q$ and $\boldsymbol{v}_{q,s} = \boldsymbol{F}\boldsymbol{V}_s\tilde{\boldsymbol{\beta}}_q$. The second relation holds because Γ_s is smooth (memory 13.8 ns): $\mathrm{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'$ stays delay-limited, so it lies in the same DPSS span with a state-specific coefficient matrix $\boldsymbol{V}_s$, while $\tilde{\boldsymbol{\beta}}_q$ is shared by the two states. With G distinct patterns (groups $\mathcal{G}_g$) and rank r, (4) becomes

$$\boldsymbol{H} = \sum_{g=1}^{G}\sum_{s=1}^{2} \boldsymbol{T}_{g,s} \circ \big((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^{\mathsf{T}}\big), \tag{6}$$

Algorithm 1 Timing-only low-rank estimation

Input: pilots $\{\boldsymbol{y}_p, \boldsymbol{s}_p\}$; kernels $\boldsymbol{T}_{g,s}$; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; rank r; iterations I

1: $\boldsymbol{G}_{g,s,p} \leftarrow \boldsymbol{T}_{g,s}\,\mathrm{diag}(\boldsymbol{s}_p)\boldsymbol{F}$ by FFT
2: $[\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{C}}_2] \leftarrow$ LMMSE of the linear model; rank-r SVD $\rightarrow \boldsymbol{U} = \boldsymbol{Y}_r\boldsymbol{\Sigma}_r^{1/2}$, $[\boldsymbol{V}_1; \boldsymbol{V}_2] = \boldsymbol{W}_r^*\boldsymbol{\Sigma}_r^{1/2}$, $\boldsymbol{\Lambda}_g = \boldsymbol{I}$
3: **for** $i = 1, \dots, I$ **do**
4: $\quad\{\boldsymbol{\Lambda}_g\} \leftarrow$ ridge least squares on (7); $\boldsymbol{U} \leftarrow$ (8); $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ (8)

Output: factors of (6) ($\boldsymbol{H}$ is not formed)

with $\boldsymbol{U}, \boldsymbol{V}_s \in \mathbb{C}^{L\times r}$ and $\boldsymbol{\Lambda}_g = \sum_{q\in\mathcal{G}_g} \tilde{\boldsymbol{\alpha}}_q\tilde{\boldsymbol{\beta}}_q^{\mathsf{T}} \in \mathbb{C}^{r\times r}$. The number of unknowns falls from up to M^2 to $3Lr + Gr^2$. An unswitched direct path is the special case of equal input responses in both states (since $\boldsymbol{T}_{q,1} + \boldsymbol{T}_{q,2} = \boldsymbol{I}$), so (6) contains it without change.

B. Timing-only estimation (Algorithm 1)

Let pilot symbol p be $\boldsymbol{s}_p$, with received $\boldsymbol{y}_p$, and let $\boldsymbol{G}_{g,s,p} = \boldsymbol{T}_{g,s}\,\mathrm{diag}(\boldsymbol{s}_p)\boldsymbol{F}$, computed once by FFT. By Corollary 4,

$$\boldsymbol{H}\boldsymbol{s}_p = \sum_{g,s} \big[(\boldsymbol{F}\boldsymbol{U}\boldsymbol{\Lambda}_g) \circ (\boldsymbol{G}_{g,s,p}\boldsymbol{V}_s)\big]\mathbf{1}_r, \tag{7}$$

which is linear in $\boldsymbol{U}$, in $[\boldsymbol{V}_1; \boldsymbol{V}_2]$ and in $\{\boldsymbol{\Lambda}_g\}$ when the other two are fixed. For each block $\boldsymbol{\theta}$ the N_p stacked pilots give $\boldsymbol{y} = \boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \boldsymbol{w}$. With the prior $\boldsymbol{\theta} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{P})$, $\boldsymbol{P} = \mathrm{diag}(\boldsymbol{p})$, the columns of $\boldsymbol{U}$ and $\boldsymbol{V}_s$ carry variances $\varsigma\boldsymbol{\lambda}$, with the scale ς set by moment matching on $\|\boldsymbol{y}\|^2$. The LMMSE update

$$\hat{\boldsymbol{\theta}} = \boldsymbol{P}^{1/2}\big(\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{P}^{1/2} + \sigma^2\boldsymbol{I}\big)^{-1}\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{y} \tag{8}$$

stays well posed when some prior variances are tiny. Alternating (8) over the blocks is non-convex, so the initialization matters. Ignoring the coupling between groups gives a *linear* model, $\boldsymbol{H} \approx \sum_s \bar{\boldsymbol{T}}_s \circ (\boldsymbol{F}\boldsymbol{C}_s\boldsymbol{F}^{\mathsf{T}})$ with $\bar{\boldsymbol{T}}_s = \sum_g \boldsymbol{T}_{g,s}$, which is exact for $G = 1$. Its LMMSE estimate with prior $\varsigma(\boldsymbol{\lambda}\otimes\boldsymbol{\lambda})$, followed by a rank-$r$ SVD $[\hat{\boldsymbol{C}}_1\ \hat{\boldsymbol{C}}_2] \approx \boldsymbol{Y}\boldsymbol{\Sigma}\boldsymbol{W}^{\mathsf{H}}$, initializes the factors. From this initialization ALS converges in one or two iterations; random initializations end 3–5 times worse.

Why two pilot symbols can suffice. At $M = 256$ with $L = 16$ and $r = 8$ there are 448 unknowns against 512 observations from two pilot symbols, yet the problem is well determined because the prior concentrates each hop on about $M\Delta f\tau_{\max} + 1$ dimensions. We measured the effective degrees of freedom of the linear stage, the trace of its LMMSE hat matrix: 19–27 at 20 dB and 23–33 at 30 dB for 1–8 pilot symbols, against 256–2048 observations. With two pilot symbols the achieved coupling-matrix error is 5–6 dB above the Bayesian bound under the assumed prior (-26.4 vs -32.4 dB at 20 dB). The gap is not estimator suboptimality: real channels do not follow the uniform-delay prior. An earlier version kept the linear estimate whenever it fitted the pilots better. On fresh 3GPP channels this rule was an overfitting trap (chosen in 37–57% of channels at 30 dB, worse rate), so Algorithm 1 always returns the low-rank estimate.

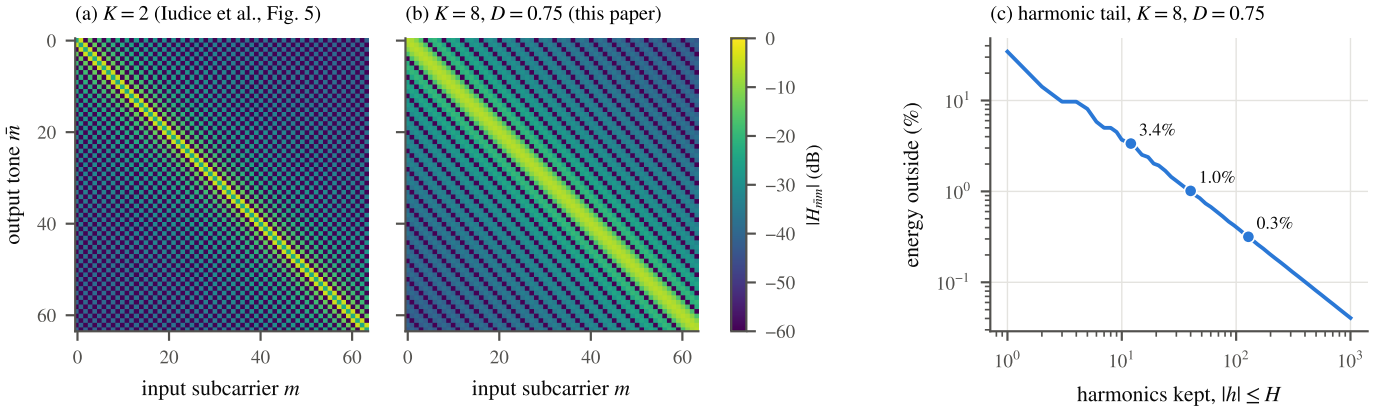


Fig. 3. Coupling-matrix magnitude for one element with ideal single-tap channels, $M = 64$, OpenRIS cell: (a) our implementation of (2) for the setting of [4, Fig. 5] ($K = 2$), reproducing it; (b) the setting of this paper ($K = 8$, $D = 0.75$). (c) Fraction of the coupling energy outside $|h| \le H$ (Corollary 3).

C. Blind clock offset and delay reference (Algorithm 2)

By Proposition 2 a candidate offset d replaces each kernel by $\boldsymbol{D}_d \boldsymbol{T}_{g,s} \boldsymbol{D}_d^*$, and a window start t_0 replaces $\boldsymbol{F}$ by $\boldsymbol{F}_{t_0} = \mathrm{diag}(e^{-j2\pi m \Delta f t_0})\boldsymbol{F}$. For each candidate the linear stage is a Gaussian model, $\boldsymbol{y} = \boldsymbol{\Phi c} + \boldsymbol{w}$ with $\boldsymbol{c} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{P})$, and its log-evidence is, up to a constant,

$$\mathcal{E}(d, t_0) = -\boldsymbol{y}^{\mathsf{H}} \boldsymbol{C}^{-1} \boldsymbol{y} - \log\det \boldsymbol{C}, \qquad \boldsymbol{C} = \boldsymbol{\Phi P \Phi}^{\mathsf{H}} + \sigma^2 \boldsymbol{I}. \tag{9}$$

With the Cholesky factor $\boldsymbol{R}^{\mathsf{H}}\boldsymbol{R} = \sigma^{-2}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{\Phi}\boldsymbol{P}^{1/2} + \boldsymbol{I}$ and the Woodbury identity, $\log\det \boldsymbol{C} = MN_p \log\sigma^2 + 2\sum_i \log R_{ii}$ and $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}^{-1}\boldsymbol{y} = \sigma^{-2}\|\boldsymbol{y}\|^2 - \sigma^{-4}\|\boldsymbol{R}^{-\mathsf{H}}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{y}\|^2$, so the evidence costs one $2L^2$-dimensional factorization per candidate.

Why the evidence. By (5), a wrong offset can still explain the pilots, through a delayed user hop and an advanced BS hop. The pilot residual is therefore nearly flat in d for physical offsets below about 100 ns. For such offsets, the residual of the advanced response after projection onto the delay basis is only $2.7 \cdot 10^{-4}$ at 100 ns (relative). A wrong offset thus *overfits*: it fits through improbable shifted responses. The log-determinant in (9) penalizes exactly this. With two held-out pilot symbols, the offset that minimizes the held-out residual agrees with the evidence maximizer within 0.005 slot in 93% of channels. The window start must be estimated too. After synchronization the first arrivals of the two hops sum to zero up to the offset, so a window that starts after the true reference cannot hold both hops; a synchronization error has the same effect. The evidence of different priors is directly comparable, so t_0 is chosen jointly with d.

D. Matrix-free equalization and complexity

The MMSE symbol estimate solves $(\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I})\boldsymbol{x} = \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}$. We solve it by conjugate gradients (CG), as in [10]. Each product goes through (6): for $G = 1$, the FFTs of all r columns and all data symbols form one batched call per state, the two states are summed in the frequency domain, and one batched inverse FFT follows (Algorithm 3). Unlike the BEM approximation in [10], this representation is exact.

Algorithm 2 Offset and delay reference by evidence

Input: pilots; nominal kernels; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; $\mathcal{T}_0 = \{-150, -75, 0, 75\}$ ns

1: **for** $t_0 \in \mathcal{T}_0$ **do**
2: $\quad d_0 \leftarrow \arg\max_d \mathcal{E}(d, t_0)$ on a 41-point grid over $[-\frac{1}{2}, \frac{1}{2}]$ slot
3: $\quad \hat{d}_{t_0} \leftarrow$ golden-section search of $\mathcal{E}(\cdot, t_0)$ on $d_0 \pm 0.025$ (14 steps)

Output: $(\hat{d}, \hat{t}_0) = \arg\max_{t_0} \mathcal{E}(\hat{d}_{t_0}, t_0)$; run Algorithm 1 with them

Algorithm 3 Matrix-free CG-MMSE equalization

Input: data $\boldsymbol{Y}_d$ (symbols as columns); factors of $\hat{\boldsymbol{H}}$; σ^2; $\epsilon = 3 \cdot 10^{-3}$; $i_{\max} = 50$

$\boldsymbol{B} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{Y}_d$; $\boldsymbol{X} \leftarrow \boldsymbol{0}$; $\boldsymbol{R} \leftarrow \boldsymbol{B}$; $\boldsymbol{P} \leftarrow \boldsymbol{R}$
for $i = 1, \ldots, i_{\max}$ **do** ▷ per column
$\quad \boldsymbol{Q} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}}(\hat{\boldsymbol{H}}\boldsymbol{P}) + \sigma^2\boldsymbol{P}$ ▷ batched FFT products
$\quad \alpha \leftarrow \|\boldsymbol{r}\|^2 / \boldsymbol{p}^{\mathsf{H}}\boldsymbol{q}$; $\boldsymbol{x} \leftarrow \boldsymbol{x} + \alpha\boldsymbol{p}$; $\boldsymbol{r}' \leftarrow \boldsymbol{r} - \alpha\boldsymbol{q}$
$\quad$ **if** $\|\boldsymbol{r}'\|/\|\boldsymbol{b}\| < \epsilon$ for all columns **break**
$\quad \boldsymbol{p} \leftarrow \boldsymbol{r}' + (\|\boldsymbol{r}'\|^2/\|\boldsymbol{r}\|^2)\boldsymbol{p}$; $\boldsymbol{r} \leftarrow \boldsymbol{r}'$

The stopping tolerance was chosen by a pre-registered sweep. With $\epsilon = 10^{-2}$ the symbol error rate (SER) is 1.15 times the dense solve's at $M = 3276$; with $\epsilon = 3 \cdot 10^{-3}$ it is within 2% of it at every M from 1024 to 6552, after 8–20 iterations.

Complexity (complex multiply–adds). One ALS iteration costs $\sum_{n \in \{Lr, 2Lr, Gr^2\}} (MN_p n^2 + n^3/3)$; the linear stage costs $MN_p(2L^2)^2 + (2L^2)^3/3$. For $G = 1$, $r = 8$, $N_p = 2$ and 20 iterations this totals $5.7 \cdot 10^8$ at $M = 256$ and $3.9 \cdot 10^{10}$ at the full $M = 3276$. One CG iteration costs two products of $\mathcal{O}(rM \log M)$, together about $1.7 \cdot 10^7$ operations at $M = 3276$, so a symbol costs at most $3.4 \cdot 10^8$ with 20 iterations. The dense alternative forms $\hat{\boldsymbol{H}}$ (172 MB at $M = 3276$, 687 MB at $M = 6552$) and factors $\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I}$ once per estimate ($4.7 \cdot 10^{10}$ operations), after which each symbol costs $\mathcal{O}(M^2)$. Measured on an idle 4-thread CPU

($\epsilon = 3\cdot10^{-3}$, $M = 3276$), dense equalization of 1, 12, 40 and 120 symbols per estimate took 2.51, 2.54, 2.58 and 2.78 s and CG 0.23, 2.51, 8.2 and 23.8 s, with CG's SER within 2% of dense. The break-even is thus about 12 symbols, not the 140 the operation count suggests, because the dense factorization runs as level-3 BLAS. CG is the faster choice for the 6–15-symbol blocks selected at 30 and 120 km/h (Sec. V-F); dense is 3–9× faster for slow channels. CG's memory is linear in M: at $M = 6552$ it equalized 120 symbols in 51 s without the 687 MB matrix. Warm-starting a refresh from the previous factors (two ALS iterations, no linear stage) took 1.4 s against 12.5 s from scratch, with the same SER.

V. Evaluation

Setup. We use the n78 carrier with $\Delta f = 30$ kHz, $T_{\mathrm{cp}} = 2.34\,\mu$s, $Q = 16$ elements (4×4, $\lambda/2$), $K = 8$, $D = 0.75$, identical patterns unless stated, and the OpenRIS cell (1). Comparisons use $M = 256$ (central 128 subcarriers scored); full-band results use $M = 3276$. Channels:

- a *shared-path* channel: per hop, one line-of-sight and five scattered paths, 30 ns rms, Rician factor 10 dB;
- 3GPP TDL-A/C/D [14] at 100 ns, with taps from random directions.

Pilots are QPSK pilot symbols. The estimator uses $\tau_{\max} = 0.3\,\mu$s, $L = 16$, $r = 8$ on the shared-path channel and $\tau_{\max} = 1\,\mu$s, $L = \lceil M\Delta f\tau_{\max}\rceil + 10$, $r = 16$ on TDL. These hyperparameters were frozen before the tests below. The metric is the achievable rate of the MMSE filter $\boldsymbol{W}$ built from the estimate and applied to the *true* channel: with $\boldsymbol{G} = \boldsymbol{W}\boldsymbol{H}$, $\mathrm{SINR}_m = |G_{mm}|^2/(\sum_{n\neq m}|G_{mn}|^2 + \sigma^2\|\boldsymbol{w}_m\|^2)$, and $R = \mathrm{mean}_m \log_2(1 + \mathrm{SINR}_m)$. The *loss* is R with perfect channel state information (CSI) minus R with the estimate. Each test was pre-registered with its pass criterion and uses 30 independent channels per point; intervals are paired 95% t-intervals.

Baselines. All baselines get the same pilots, the same prior family and, for fairness, three equalizers each: naive MMSE, robust MMSE with the estimation-error power taken from the pilot residual, and robust MMSE with the oracle error power. Figures show the residual-based robust MMSE, which is usable in practice; paired comparisons take, per channel, the best over all baselines and all three equalizers.

- *One-tap*: today's OFDM receiver.
- *Ideal-element LMMSE*: the strongest estimator that assumes the textbook element, $\boldsymbol{H} = (\boldsymbol{T}_1 - \boldsymbol{T}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$ with our prior.
- *CE-BEM* [6]: $\boldsymbol{H} = \sum_{|h|\le H_b} \boldsymbol{S}_h \,\mathrm{diag}(\boldsymbol{F}_2\boldsymbol{c}_h)$ with up to all $2M - 1$ harmonics, solved matrix-free (equal to the dense solver to $7\cdot10^{-9}$); with all harmonics it is run from eight pilot symbols.
- *OMP* [15] over a harmonic×delay dictionary.
- *Learned-prior LMMSE*, the learning-based baseline: a principal-component basis of $\mathrm{vec}(\boldsymbol{H})$ and its eigenvalues learned from N_{tr} true channels form a Gaussian prior, and the LMMSE estimate follows from the pilots. It is the Bayes-optimal linear estimator for the learned statistics, which a network trained for minimum mean squared error approaches for near-Gaussian channels, and it learns the element non-ideality from data (Sec. V-B).

Two references are not practical receivers but locate the methods between the extremes: the *exact-element LMMSE*, which is the ideal-element LMMSE given the true responses, $\boldsymbol{H} = (\boldsymbol{T}_1\,\mathrm{diag}\,\boldsymbol{\Gamma}_1 + \boldsymbol{T}_2\,\mathrm{diag}\,\boldsymbol{\Gamma}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$, and the same estimator with both resonances shifted by 5–20 MHz (a mismatched model).

A. Comparison with the strongest baselines

Below 15 dB the receivers that know the element model are slightly better than ours (by ≤ 0.02 bit/s/Hz here, ≤ 0.06 on TDL-C), since a linear LMMSE with the right structure needs no nonlinear fit. From 20 dB the order reverses and the gaps grow: at 20/30 dB ours loses 0.08/0.17 bit/s/Hz, the exact-element LMMSE 0.10/0.36, the 20 MHz-mismatched model 0.13/0.56, CE-BEM with 16 times our pilots 0.44/0.57, the ideal-element LMMSE 0.50/2.32 and OMP 4.1/7.5. Ours beats the exact-element LMMSE by having fewer unknowns (rank $r = 8$ against $L^2 = 256$), not by knowing more. On TDL-C ($N_p = 4$) the 20/30 dB losses are 0.20/0.74 (ours), 0.26/1.02 (exact element), 0.63/2.55 (ideal element) and 1.21/1.64 (CE-BEM, $N_p = 32$). Over 10–40 dB (Fig. 4, a separate set of channels), ours has the lowest NMSE at every SNR on the shared path (-23.7 to -48.4 dB) and stays below 0.3 bit/s/Hz up to 40 dB, where the exact-element LMMSE loses 0.46, CE-BEM ($N_p = 32$) 0.67 and the ideal-element LMMSE 5.4, whose NMSE floors at -24 dB. Its BER is within a factor of about two of perfect CSI up to 30 dB for both modulations. On TDL-C every estimator floors at high SNR: ours crosses 0.3 bit/s/Hz near 22 dB and loses 2.3 at 40 dB, and from 35 dB CE-BEM with eight times our pilots has the lower BER ($2.8\cdot10^{-4}$ against $7.1\cdot10^{-4}$ for 16-QAM at 40 dB) at a similar rate loss (2.5 against 2.3).

Fig. 5 shows that the literature's estimator is not a weak baseline. As $\boldsymbol{H}$ is linear in Γ_s, the ideal and the physical element can be blended. With ideal elements ($\alpha = 0$) the ideal-element LMMSE is the exact model and matches ours (0.053 against 0.078 at 20 dB, within the intervals). As the non-ideality grows, its loss rises past 0.3 bit/s/Hz at $\alpha = 0.5$ (20 dB) and $\alpha = 0.25$ (30 dB), while ours stays at 0.07–0.08 (0.15–0.18 at 30 dB). An approximate element model is nearly as good as the exact one (the 5–20 MHz mismatch costs ≤ 0.4 dB of NMSE at 20 dB); assuming the ideal switch is what fails (Corollary 1).

Ours (Fig. 6) is the only estimator within 0.3 bit/s/Hz with at most 32 pilot symbols: two on the shared-path channel (0.07 ± 0.02 / 0.11 ± 0.03 at 20/30 dB), four on TDL-C. Its NMSE falls about 3 dB per doubling of N_p (-29.3 to -45.4 dB, N_p = 1–32, 20 dB); the ideal-element LMMSE saturates at -24.6 dB and CE-BEM needs $N_p \ge 8$. Paired per channel against the best model-based baseline on that channel (minimum over those baselines and all equalizers), ours gains 0.41 ± 0.02 / 2.16 ± 0.04 bit/s/Hz at $N_p = 2$ on the shared-path

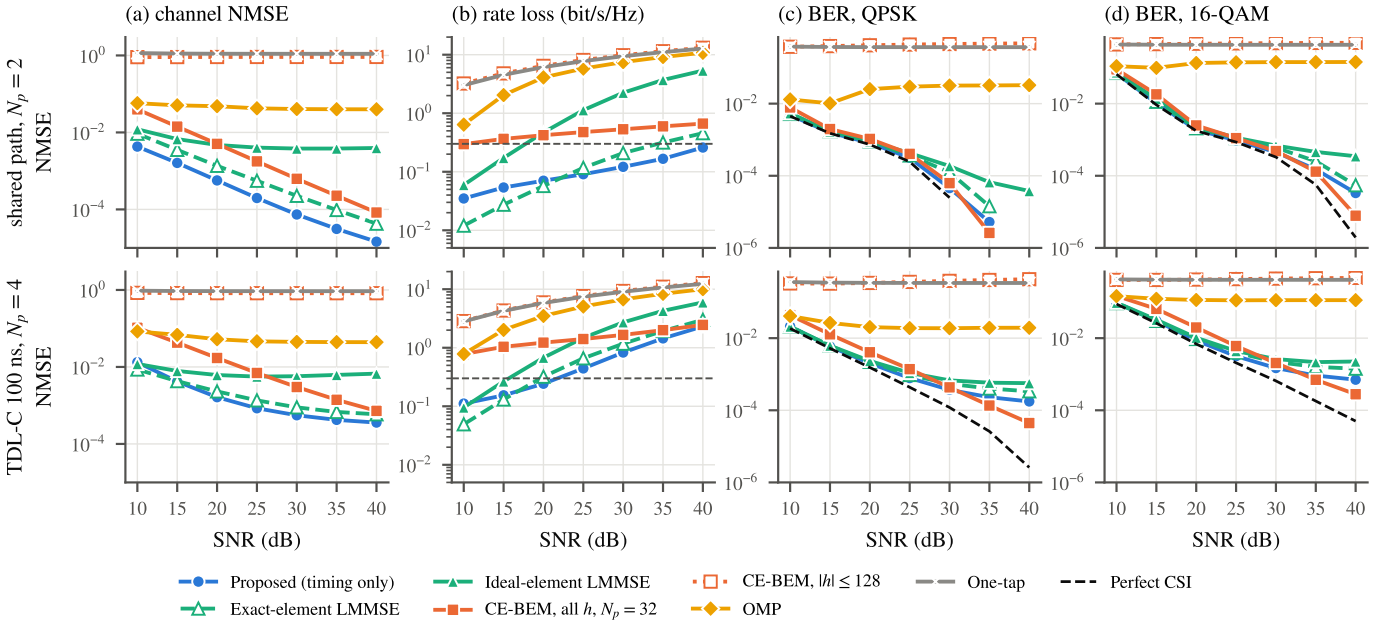


Fig. 4. Estimators versus SNR in the format of the CE-BEM paper [6] (Figs. 10–11 there): 10–40 dB, 30 new channels per point; top: shared path, $N_p = 2$; bottom: TDL-C, $N_p = 4$ (CE-BEM with all harmonics: $N_p = 32$). (a) NMSE, (b) loss to perfect CSI with the residual-based robust MMSE (dashed: 0.3 bit/s/Hz), (c), (d) uncoded BER after unbiased MMSE ($7.7 \cdot 10^5$ / $1.5 \cdot 10^6$ bits per point; points with no errors are omitted).

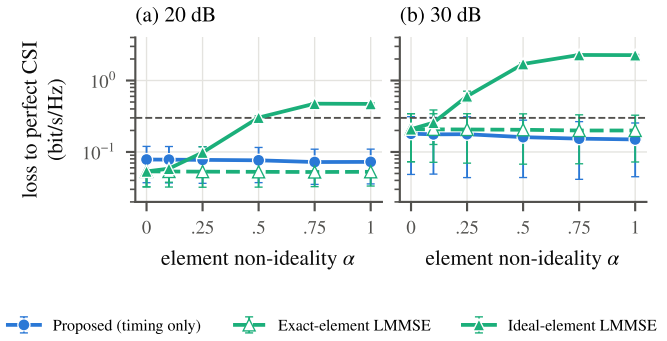


Fig. 5. Loss versus element non-ideality, $\boldsymbol{H}_\alpha = (1-\alpha)\boldsymbol{H}_{\text{ideal}} + \alpha\boldsymbol{H}_{\text{OpenRIS}}$ (shared path, $N_p = 2$, 30 channels, 95% intervals). CE-BEM ($|h| \le 128$) and one-tap lose more than 6 bit/s/Hz at every α.

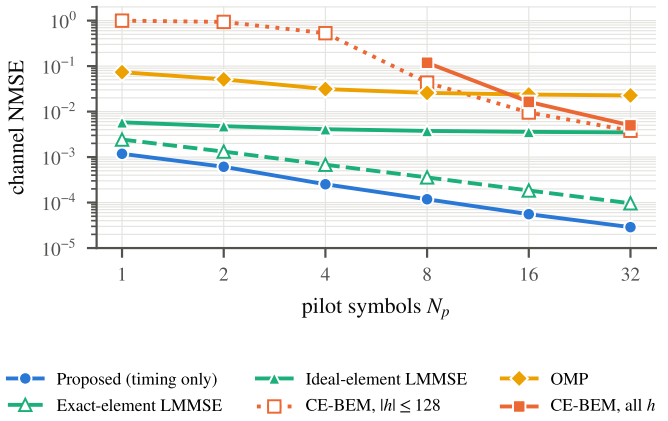


Fig. 6. NMSE versus pilot symbols, shared-path channel, 20 dB, 30 channels. CE-BEM with all harmonics is run from $N_p = 8$.

channel and 0.30 ± 0.06 / 1.14 ± 0.25 on TDL-C; at every N_p from 2 to 32 the gain is 0.3–0.4 at 20 dB and 0.5–2.2 at 30 dB, with every interval above zero. A detector that estimates G_{mm} adds at most 0.15 to any loss. One pre-registered sanity check failed: truncated CE-BEM loses rate from 20 to 30 dB even with the oracle robust equalizer (its error sits on the missing harmonics).

B. Learning-based baseline and a self-trained prior

Table II answers whether learning would do better. It would, if trained on the right channels: the learned-prior LMMSE beats ours on every family when trained on the same distribution, already with $N_{\text{tr}} = 100$ (0.006 against 0.070 at 20 dB on the shared path with $N_{\text{tr}} = 1000$). The reason is that our simulated families are low-dimensional. At $M = 256$ the band is 7.7 MHz, so a 30 ns channel with a strong line of sight is nearly flat (the 1000 shared-path channels span 6 principal components, with mean correlation 0.92), and TDL has fixed tap delays (50 components). A prior learned on one family transfers poorly to a different delay spread or Rician factor: trained on the 30 ns family and tested at 100 ns it is worse than ours by 0.22 ± 0.17 / 0.95 ± 0.46 bit/s/Hz, and trained on TDL-C it ties ours on TDL-D at 20 dB and is worse by 0.37 ± 0.20 at 30 dB; between the similar TDL-A and TDL-C it still gains 0.06 ± 0.03 at 20 dB. Our estimator needs neither training channels nor an element model, and it can supply the training data itself: a prior learned from its 8-pilot estimates of 300 channels beats it by 0.042 ± 0.012 (shared path) and

TABLE II

LOSS TO PERFECT CSI (BIT/S/HZ) AT 20/30 DB AGAINST THE LEARNED-PRIOR LMMSE ($N_{\mathrm{tr}} = 300$, 30 TEST CHANNELS; $N_p = 2$ SHARED PATH, 4 TDL). SHIFTED: TRAINED ON THE FAMILY IN BRACKETS. SELF-TRAINED: PRIOR LEARNED FROM OUR OWN ESTIMATES OF 300 CHANNELS WITH 8 PILOT SYMBOLS AT 20 DB.

Test channels	Ours	Learned	Learned, shifted	Self-trained
Shared, 30 ns, K 10 dB	.071/.143	.007/.031	–	.029/.118
Shared, 100 ns, K 3 dB	.124/.468	.054/.247	.346/1.42 (30 ns)	–
TDL-C	.185/.668	.058/.213	–	.078/.362
TDL-A	.178/.726	.047/.184	.121/.611 (C)	–
TDL-D	.483/1.68	.043/.204	.445/2.05 (C)	–

TABLE III

ESTIMATOR UNCHANGED ON DATA VIOLATING ITS MODEL: RELATIVE CHANGE OF $\boldsymbol{H}$ AND EXTRA LOSS (BIT/S/HZ, PAIRED, 30 CHANNELS).

Violation	$\Delta\boldsymbol{H}$	20 dB	30 dB
Time-domain truth	0.006	-0.004 ± 0.005	$+0.006 \pm 0.009$
Switching transients	0.001	$+0.001 \pm 0.001$	$+0.003 \pm 0.006$
State-dep. output, $\epsilon = 0.2$	0.40	≤ 0.004	$+0.034$
Joint-state coupl., $\kappa = 0.08$	0.29	≤ 0.002	$+0.015$
Resonance spread 10 MHz	0.18	$+0.003 \pm 0.008$	$+0.046 \pm 0.044$
Beam squint, $Q = 256$	0.016	≤ 0.001	-0.002

0.107 ± 0.027 bit/s/Hz (TDL-C) at 20 dB. Because such a prior is collected in place, it matches the deployment distribution by construction; we offer it as an optional second stage.

C. No element model: robustness beyond the generating model

A receiver tested on data from its own model can look better than it is. Table III runs the estimator *unchanged* on data that violate its structure, paired with the same channel without the change. In every case the extra loss stays within 0.05 bit/s/Hz, which passes the pre-registered bar of 0.3. Element-to-element resonance spread of 5–20 MHz costs 0.06–0.15 bit/s/Hz. A six-parameter fit of the element model is not better than timing-only ($+0.08 \pm 0.05$ worse at 30 dB) and fails with element spread ($+0.73 \pm 0.43$). On a 16-element surface the linear first stage provides most of the gain (loss 0.10/0.23 at $N_p = 2$); the low-rank step and one decision-directed pass reduce it to 0.07/0.11 and 0.02/0.03. The low-rank property itself matters most for large surfaces: the effective rank of the true element-domain matrix (99% energy, CDL-C, 98 MHz, beam squint) is 5, 9, 13 and 16 for $Q = 16$, 64, 256 and 1024.

D. Timing

Table IV tests Algorithm 2 with physical bulk delays ($\tau_b \sim \mathcal{U}(50, 300)$ ns, $\tau_u \sim \mathcal{U}(30, 300)$ ns). A BS-clock-locked RIS needs the search, and with the search the rate is within noise of known timing. An incident-signal-locked RIS needs no search, and running it anyway costs nothing significant. A synchronization error of up to ± 150 ns breaks a fixed delay window, but not the jointly chosen one. Is the evidence necessary? On a fine grid the pilot residual of the full low-rank estimate also finds the offset with two or more pilot symbols (not significantly worse). With a single pilot symbol it overfits: it is worse by 1.49 ± 0.56 / 2.83 ± 1.19 bit/s/Hz, while the evidence stays within 0.03 of known timing at 20 dB. The evidence is also 5–58 times cheaper per search (3.2 vs 17.2 s at $M = 256$; 11.7 vs 674 s on TDL-C), because it needs no low-rank estimate per candidate. Random per-edge jitter cannot be estimated. It must stay below about 4 ns rms at 30 dB (20 ns at 20 dB), whereas ordinary clocks have picosecond jitter.

TABLE IV

TIMING: LOSSES AND PAIRED DIFFERENCES (BIT/S/HZ, 30 CHANNELS).

	20 dB	30 dB
BS-clock RIS, offset ignored	1.62	4.23
Alg. 2 minus known offset	$+0.011 \pm 0.019$	$+0.033 \pm 0.066$
Incident-locked: Alg. 2 minus none	$+0.001 \pm 0.007$	$+0.040 \pm 0.082$
Sync error ≤ 150 ns, fixed window	0.22	1.16
window chosen by evidence	0.05 ± 0.05	0.05 ± 0.08
$N_p = 1$: residual minus evidence	1.49 ± 0.56	2.83 ± 1.19

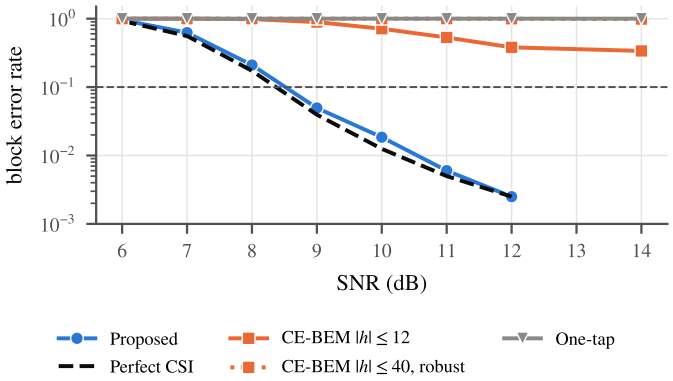


Fig. 7. Coded BLER, $N_p = 2$, 2000 blocks per point. Gap to perfect CSI at 10% BLER: 0.14 dB [0.09, 0.20].

E. Full band and coded link

At the full band ($M = 3276$, 30 channels) the SER with two pilot symbols is 1.10 [1.06, 1.15] times the perfect-CSI SER at 20 dB with 16-QAM and 1.37 [1.18, 1.86] times at 28 dB with 64-QAM (95% clustered-bootstrap intervals). One decision-directed pass brings these to 1.05 and 1.12. CE-BEM ($|h| \leq 8$) is 18 and 32 times worse; the one-tap receiver's SER is 0.84 and 0.96. For the coded link (a rate-1/2 convolutional code, 16-QAM, max-log LLRs from each receiver's own estimate, and 2000 blocks per point) the proposed receiver (Fig. 7) crosses 10% block error rate (BLER) at 8.51 dB against 8.37 dB with perfect CSI, a gap of 0.14 dB [0.09, 0.20]. One-tap never decodes, and truncated CE-BEM floors at 34%.

F. Mobility and pilot overhead

When the user moves, only the RIS–user hop changes. Each path rotates by $e^{j2\pi\nu_\ell n T_s}$ between symbols and also mixes subcarriers within a symbol. That mixing is -60, -47 and -34 dB below the signal at 3, 30 and 120 km/h, so the main effect is the ageing of the estimate. With N_p pilot symbols every N symbols, the net rate is $R_{\mathrm{net}} = \frac{1}{N}\sum_{n=N_p}^{N-1} R(n)$, where $R(n)$ uses the estimate on symbol n with one common phase corrected per symbol, as 5G phase-tracking pilots do. Table V

TABLE V
NET THROUGHPUT AFTER PILOT OVERHEAD (BIT/S/HZ), (N_p, N) SELECTED ON 20 CHANNELS AND REPORTED ON 20 DISJOINT CHANNELS; OURS − CE-BEM PAIRED, 95% INTERVAL.

		20 dB			30 dB		
Channel	km/h	ours	BEM	diff.	ours	BEM	diff.
Shared	3	6.24	5.24	1.01 ± 0.14	8.99	7.34	1.65 ± 0.34
Shared	30	5.97	4.40	1.57 ± 0.30	8.56	5.42	3.14 ± 0.40
Shared	120	5.26	3.15	2.11 ± 0.30	7.30	3.38	3.92 ± 0.53
TDL-A	3	5.38	3.71	1.67 ± 0.17	7.37	4.88	2.49 ± 0.21
TDL-A	30	4.56	1.60	2.97 ± 0.13	5.51	1.68	3.84 ± 0.27
TDL-A	120	3.81	1.08	2.73 ± 0.28	4.35	1.07	3.27 ± 0.58
TDL-C	3	5.34	3.61	1.74 ± 0.16	7.23	4.81	2.41 ± 0.27
TDL-C	30	4.53	1.65	2.88 ± 0.29	5.53	1.81	3.72 ± 0.42
TDL-C	120	3.88	0.75	3.13 ± 0.31	4.61	0.74	3.88 ± 0.54
TDL-D	3	5.17	3.61	1.56 ± 0.17	6.80	4.75	2.05 ± 0.41
TDL-D	30	4.65	2.17	2.48 ± 0.25	5.85	2.39	3.46 ± 0.42
TDL-D	120	3.92	1.15	2.77 ± 0.31	4.68	1.15	3.53 ± 0.69

selects (N_p, N) and, for CE-BEM, the harmonic range on 20 channels, and reports them on 20 *other* channels. Ours beats the best CE-BEM in all 24 cases, by 1.0–3.9 bit/s/Hz, every interval above zero (CE-BEM selects 16–32 pilot symbols). Selecting on the reported channels changes the numbers by at most 0.13. Net of pilots, ours keeps 75–93% of the perfect-CSI rate on the shared path and 73–88% on TDL at 3 km/h, but only 46–66% on TDL at 120 km/h: with a 30-symbol coherence time the selected design spends one pilot symbol in every 6. Slow channels select two (shared) or four (TDL) pilot symbols per 40–120 symbols. High mobility is thus a limit of pilot-based estimation, not of the coupling model. Two decision-directed trackers (LMMSE and Kalman) were worse than reusing the pilot estimate, because wrong decisions accumulate after about ten symbols.

G. Hardware, deployment and the cost of time modulation

Impairments. Residual CFO must stay below about 1% of the spacing (0.10/0.33 bit/s/Hz at 20/30 dB), and phase noise below about 100 Hz linewidth or be tracked (0.16/0.33 at 100 Hz). Rise times up to 42 ns need no calibration (≤ 0.011 at 30 dB); longer ones are calibrated or fitted blindly from the pilots (within 0.03 slot). Rise time also sets the emission: ideal edges give an adjacent-channel leakage ratio of 36.8 dB, below the 45 dB required of a base station [16], and 0.05-slot edges give 52.7 dB. Slow edges shrink our lead: at a 0.3-slot ramp CE-BEM with eight pilot symbols reaches 8.2 against our 8.56 bit/s/Hz with two.

Deployment. A direct path needs no change ($T_{q,1} + T_{q,2} = \boldsymbol{I}$; 5.28/8.13 against 5.39/8.49 bit/s/Hz with perfect CSI), nor do two panels with different patterns (5.54/8.67 against 5.64/8.86). An unknown OFDM interferer at 0–20 dB SIR costs 0.03–0.17; 3GPP CDL channels match TDL within 0.07.

Cost of time modulation. At equal transmit power and 20 dB, a TM-RIS with a pattern designed for the user gives 0.68/1.98/3.15 bit/s/Hz with one-tap for $D = 0.625/0.75/0.875$, 2.98/4.06/5.35 with ours, and 3.08/4.18/5.50 with perfect CSI; a static RIS optimized for the user gives 6.66. Time modulation is thus not a single-user rate gain but is used for security, harmonic steering or sensing, and our receiver recovers its cost to within 0.1–0.2 bit/s/Hz of perfect CSI. An eavesdropper with generic CE-BEM still obtains 1.8–2.2 bit/s/Hz (2.5–3.0 knowing the pattern), so the pattern must be kept secret.

VI. CONCLUSION

Grouping the harmonic expansion of a dispersive TM-RIS by element state separates the coupling exactly into a part the receiver knows, the switching timing, and a part it learns, the propagation together with the element dispersion. A receiver built on this separation needs no element model. It estimates the coupling from two to eight pilot symbols, finds the RIS clock offset by itself, and equalizes without forming the coupling matrix. In pre-registered tests at 20–30 dB it beats every estimator that needs no training data, on every channel and pilot count; with ideal elements the literature's estimator matches it and falls behind only as the elements become less ideal. A prior learned from channels of the deployment distribution is better, a prior learned elsewhere can be much worse, and a prior learned from our own estimates keeps most of the gain without ground truth. It decodes within 0.14 dB of perfect CSI. Its limits are stated above: the receiver must know every panel's pattern (A1); delay spreads whose cascade exceeds the CP are out of scope (A2); at 120 km/h on TDL channels pilot overhead and estimate ageing leave 46–66% of the perfect-CSI rate; below about 15 dB a receiver given the element model is marginally better (by ≤ 0.06 bit/s/Hz); the stated requirements on jitter, CFO, phase noise and emission apply; and the validation is by simulation with one published cell and no mutual coupling. Over-the-air validation with a real time-modulated surface is the next step.

REFERENCES

[1] L. Zhang *et al.*, "Space-time-coding digital metasurfaces," *Nat. Commun.*, vol. 9, art. 4334, 2018.

[2] Z. Xu and A. Petropulu, "Time-modulated intelligent reflecting surface for waveform security," arXiv:2310.13210, 2023.

[3] Z. Tao and A. P. Petropulu, "Secure time-modulated intelligent reflecting surface via generative flow networks," arXiv:2506.14992, 2025.

[4] I. Iudice, D. Darsena, G. Gelli, and V. Galdi, "Physically consistent modeling of dispersive time-modulated reconfigurable intelligent surfaces for wideband OFDM," arXiv:2609.18360, Sep. 2026.

[5] Z. Tao, Z. Xu, and A. Petropulu, "How secure is the time-modulated array-enabled OFDM directional modulation?" arXiv:2310.08551, 2023.

[6] Z. Tang, R. C. Cannizzaro, G. Leus, and P. Banelli, "Pilot-assisted time-varying channel estimation for OFDM systems," *IEEE Trans. Signal Process.*, vol. 55, no. 5, pp. 2226–2238, May 2007.

[7] D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case," *Bell Syst. Tech. J.*, vol. 57, no. 5, pp. 1371–1430, 1978.

[8] T. Zemen and C. F. Mecklenbräuker, "Time-variant channel estimation using discrete prolate spheroidal sequences," *IEEE Trans. Signal Process.*, vol. 53, no. 9, pp. 3597–3607, Sep. 2005.

[9] P. Wang, J. Fang, H. Duan, and H. Li, "Compressed channel estimation for intelligent reflecting surface-assisted millimeter wave systems," *IEEE Signal Process. Lett.*, vol. 27, pp. 905–909, 2020.

[10] T. Hrycak, S. Das, G. Matz, and H. G. Feichtinger, "Low complexity equalization for doubly selective channels modeled by a basis expansion," *IEEE Trans. Signal Process.*, vol. 58, no. 11, pp. 5706–5719, Nov. 2010.

[11] D. J. C. MacKay, "Bayesian interpolation," *Neural Comput.*, vol. 4, no. 3, pp. 415–447, 1992.
[12] H. Chen, R. Long, Y.-C. Liang, and R. Schober, "Blind timing estimation and signal detection for RIS-assisted symbiotic radio with imperfect symbol synchronization," in *Proc. IEEE ICC*, 2024, pp. 4155–4160.
[13] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002.
[14] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901.
[15] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.
[16] 3GPP, "NR; Base station (BS) radio transmission and reception," TS 38.104.